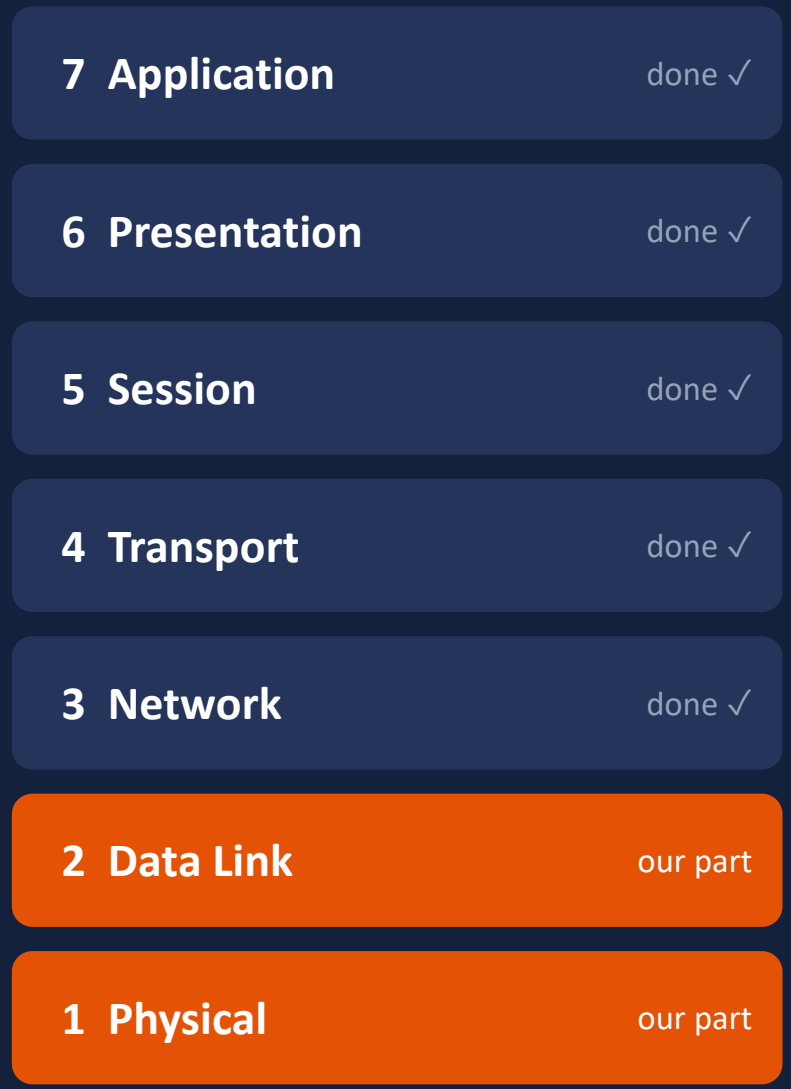
INTRODUCTION TO NETWORK SYSTEMS

From KMITL to Chiang Mai

The journey so far, and welcome to the Link Layer

Week 11

Dr. Thananop Thongthavorn
School of Information Technology, KMITL



Our five classes

Layer 2 in Classes 1–3, Layer 1 in Classes 4–5



TODAY

Class 1

LIVE Layer 2

Recap of L7–3, link layer basics, MAC addresses, ARP

Human ARP



Class 2

VIDEO Layer 2

Error detection: parity, checksum, CRC

Worksheet



Class 3

LIVE Layer 2

Multiple access, Ethernet, switches, VLANs, Wi-Fi

Talking Game, Human Switch



Class 4

LIVE Layer 1

Transmission media, signals, line coding

Cable show-and-tell, flashlight bits



Class 5

VIDEO Layer 1

Bandwidth, Shannon, modulation, multiplexing, wrap-up

Worksheet, final project

Today's plan

3 hours · 3 blocks · a 10-minute break between each

1

0:00 – 1:00

The journey so far

Recap Layers 7–3 with one story: sending a gift from KMITL to Chiang Mai

2

1:10 – 1:55

Welcome to the Link Layer

What Layer 2 does, where it lives, and why we need MAC addresses

3

2:05 – 3:00

ARP: from IP to MAC

How a device finds the MAC address it needs, plus the Human ARP activity

Keep your phone handy: we will use it for polls and one activity.

By the end of today, you can…

Six learning outcomes for Class 1

1 — Block 1

Explain the job of each OSI layer with an everyday analogy

2 — Block 1

Describe encapsulation and which devices read which layers

3 — Block 1

Explain why IP stays the same end to end while MAC changes every hop

4 — Block 2

Describe what the link layer does and where it runs

5 — Block 2

Read a MAC address and tell unicast from broadcast

6 — Block 3

Trace ARP inside a subnet and across a router

Outcome 3 is the key idea of the day.

Why split networking into layers?

The same reason a parcel company has separate teams



Divide and conquer

Each layer solves one problem: routing, ordering, or moving bits.

→ Separate teams for sorting, driving, and delivering



Clear interfaces

Each layer uses the service below and offers a service above.

→ Drivers never open boxes; you never plan truck routes



Swap one layer

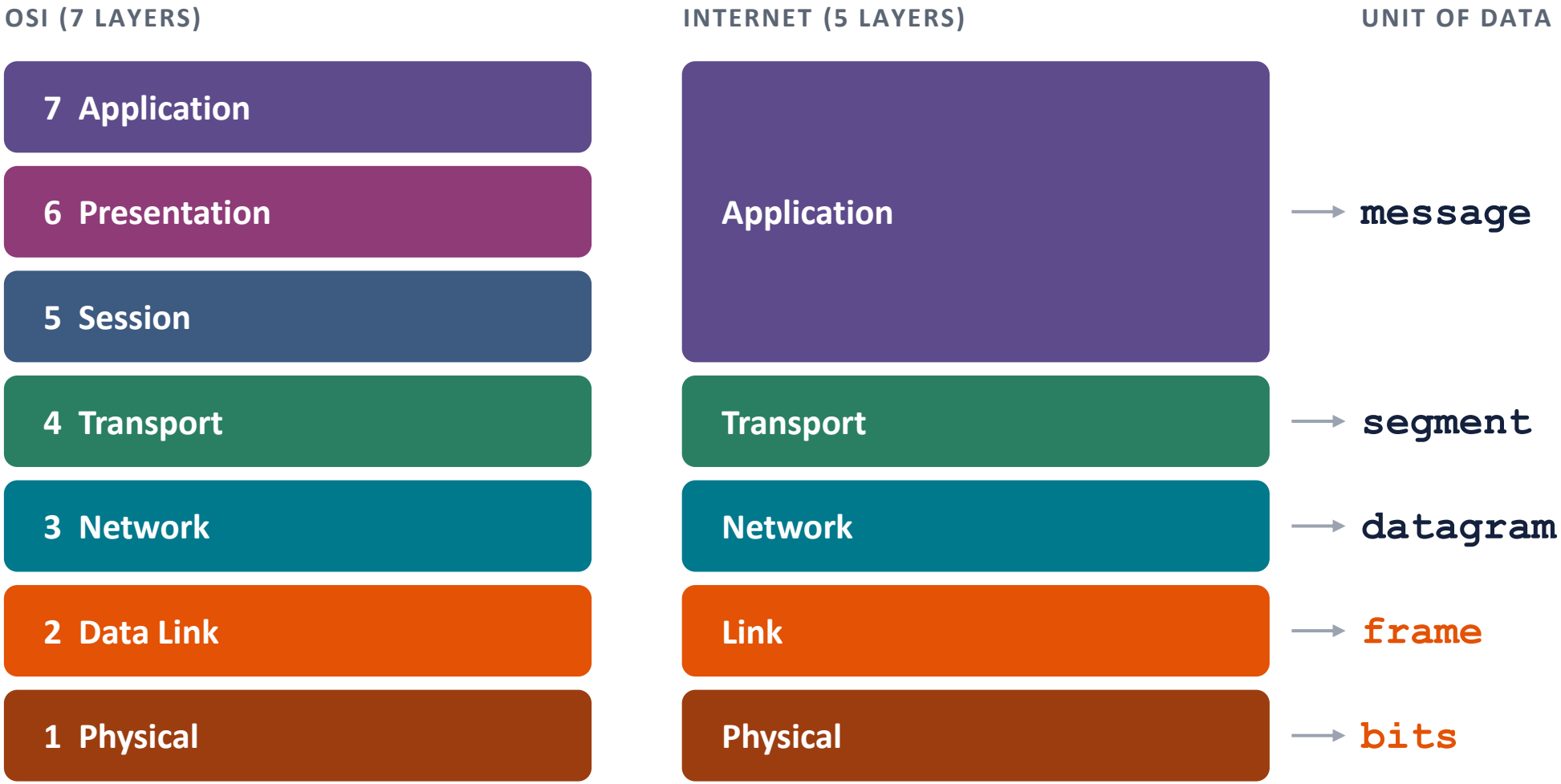
Move from Wi-Fi to Ethernet, and your browser stays the same.

→ Trucks become trains; your card stays the same

The price: every layer adds a header (overhead), and some jobs, like error checking, are repeated in several layers.

OSI model vs the Internet model

Kurose uses the five-layer Internet (TCP/IP) model



The bottom three layers are the same in both models.

The mission: a birthday gift to Chiang Mai

You are in your dorm at KMITL. Your friend lives in a condo in Chiang Mai. You send the gift with a parcel company.



Your dorm
KMITL, Ladkrabang
Your laptop

Bangkok hub
sorting center
Router

Nakhon Sawan hub
sorting center
Router

Chiang Mai hub
sorting center
Router

Friend's condo
Chiang Mai
Destination host

~700 km
by road

3
sorting hubs (routers)

4
separate truck trips (links)

One journey, seven layers

Kurose's Internet model merges Layers 5–7 into a single application layer.

LAYER		IN THE PARCEL JOURNEY	IN THE NETWORK
7	**Application**	The gift and the birthday card	HTTP, DNS
6	**Presentation**	Wrapping, vacuum-packing, locking the box	Encoding, TLS
5	**Session**	"Tell me when all 3 parcels arrive"	Session control
4	**Transport**	Tracked or untracked service, "box 1 of 3"	TCP, UDP, ports
3	**Network**	Full address, sorting hubs, condo front desk	IP, routers, NAT
2	**Data Link**	One truck trip between two neighboring hubs	**MAC, Ethernet, Wi-Fi**
1	**Physical**	The road: highway, railway, or air	**Cable, fiber, radio**

LAYER

7

Application



In our parcel journey

You choose the gift and write the birthday card. You only know your friend's name, so you look up their address in your LINE chat.

In the network

The application creates the message you actually care about. DNS turns a name like kmitl.ac.th into an IP address.

KEY TERMS

HTTP · DNS · SMTP · message

LAYER

6

Presentation



In our parcel journey

You wrap the gift, vacuum-pack the clothes so they fit in the box, and lock the box so only your friend can open it.

In the network

Data is put into a format both sides understand, compressed to save space, and encrypted to keep it private.

KEY TERMS

- encoding (UTF-8)
- compression
- encryption (TLS)

LAYER

5

Session



In our parcel journey

"I'm sending 3 parcels this week. Let me know when all of them arrive." You keep the whole exchange organized.

In the network

Opens, maintains, and closes a conversation between two applications, and can resume it after an interruption.

KEY TERMS

session setup · checkpoint · teardown

LAYER

4

Transport



In our parcel journey

You pick the service: tracked with signature, or cheap with no tracking. A big gift is split into boxes labeled "1 of 3, 2 of 3, 3 of 3". The recipient's name says who in the building gets it.

In the network

TCP is reliable and ordered, with acknowledgments. UDP is fast with no guarantees. Segmentation splits data, and port numbers deliver it to the right application.

KEY TERMS

TCP · UDP · segment · port

Tracked or untracked? TCP vs UDP

Layer 4 offers applications two kinds of service

	TCP	UDP
In our story	Tracked parcel with signature	Cheap parcel, no tracking
Setup	Handshake before sending	None: just send
Lost data	Detected and sent again	Gone
Order	Put back in order	Not guaranteed
Header size	20 bytes or more	8 bytes
Good for	Web pages, email, file transfer	DNS, video calls, online games
Well-known ports	HTTP 80 · HTTPS 443 · SSH 22	DNS 53 · DHCP 67/68

Port number = the recipient's name: which application inside the host gets the data.

LAYER

3

Network



In our parcel journey

The full address with postal code goes on the box. Each sorting hub reads only the destination and picks the next hub. The condo front desk receives parcels for the whole building and knows which room each one goes to.

In the network

IP addresses and prefixes (subnets) identify networks and hosts. Routers use forwarding tables to choose the next hop. NAT lets many private hosts share one public address.

KEY TERMS

IP · subnet · router · NAT · datagram

A subnet is like a postal code

Routers, like sorting hubs, only read the first part

PARCEL ADDRESS (LAT KRABANG)

10520
Postal code: which area

99/1
House number

IP ADDRESS 192.168.1.10/24

192.168.1
Network part: 24 bits

.10
Host part: 8 bits

What a hub (router) reads

A hub in Nakhon Sawan only looks at the postal code. A router only looks at the network part to choose the next hop. Only the last stop cares about the house or host number.

Quick check: what is the network address of 192.168.1.10/24, and how many usable hosts does a /24 have?

Bangkok to Chiang Mai, one hop at a time

Every hub reads only the postal code, then chooses the next hub. Nobody plans the whole trip.

The parcel route



≈ 700 km · 3 sorting hubs · 4 separate truck trips

The box is addressed to 10.0.0.20 — every hub looks up only the network part, 10.0.0.0/24.

1 Bangkok

hub 1 · router R1

FORWARDING TABLE

Destination	Next hop
10.0.0.0/24	to hub 2
192.168.1.0/24	deliver here

2 Nakhon Sawan

hub 2 · router R2

FORWARDING TABLE

Destination	Next hop
10.0.0.0/24	to hub 3
192.168.1.0/24	to hub 1

3 Chiang Mai

hub 3 · router R3

FORWARDING TABLE

Destination	Next hop
10.0.0.0/24	deliver here
192.168.1.0/24	to hub 2

No hub knows the whole route.
Each one reads the destination, finds it in its own table, and hands the parcel to the next hub. That is all a router ever does.

Postal code → next hub. Network prefix → next router. It is the same lookup, and it is why the prefix exists.

If the road to Nakhon Sawan floods, hub 1 picks a different next hop. The address on the box never changes.

NAT is the condo front desk

Many rooms, one building address

Inside the condo (private)

Laptop 1
192.168.1.10

Laptop 2
192.168.1.11



Front desk
(NAT router)

The Internet only sees

203.0.113.5

one building address for everyone inside

THE FRONT DESK'S RECORD BOOK (NAT TRANSLATION TABLE)

Inside (private IP : port)	Outside (public IP : port)
192.168.1.10 : 51000	203.0.113.5 : 40001
192.168.1.11 : 51000	203.0.113.5 : 40002

LAYER

2

Data Link



OUR PART · CLASSES 1–3

In our parcel journey

Each hop is a separate truck trip with its own trip sheet: "Bangkok hub, dock 3 → Nakhon Sawan hub, dock 1." At every hub, staff check that the box is not damaged.

In the network

Moves a frame across one link to the next node. MAC addresses identify the two ends of that link. Error detection catches bits that were flipped on the way.

KEY TERMS

frame · MAC address · ARP · switch · Ethernet · Wi-Fi

LAYER

1

Physical



OUR PART · CLASSES 4–5

In our parcel journey

The road itself: highway, railway, or air. Potholes and heavy rain cause trouble, and the speed limit caps how fast you can go.

In the network

Sends raw bits as signals over copper, fiber, or radio. Noise and attenuation damage signals, and bandwidth limits the data rate.

KEY TERMS

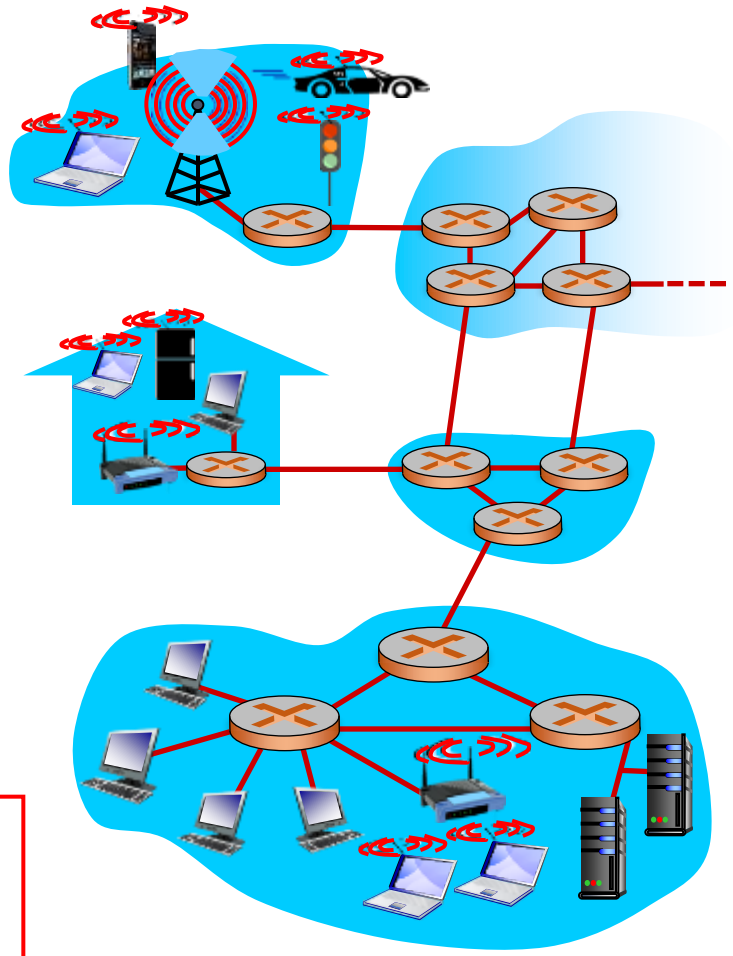
bits · signal · copper · fiber · radio · bandwidth

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Link layer: introduction

terminology:

- hosts and routers: nodes
- communication channels that connect adjacent nodes along communication path: links
  - wired links
  - wireless links
  - LANs
- layer-2 packet: frame, encapsulates datagram



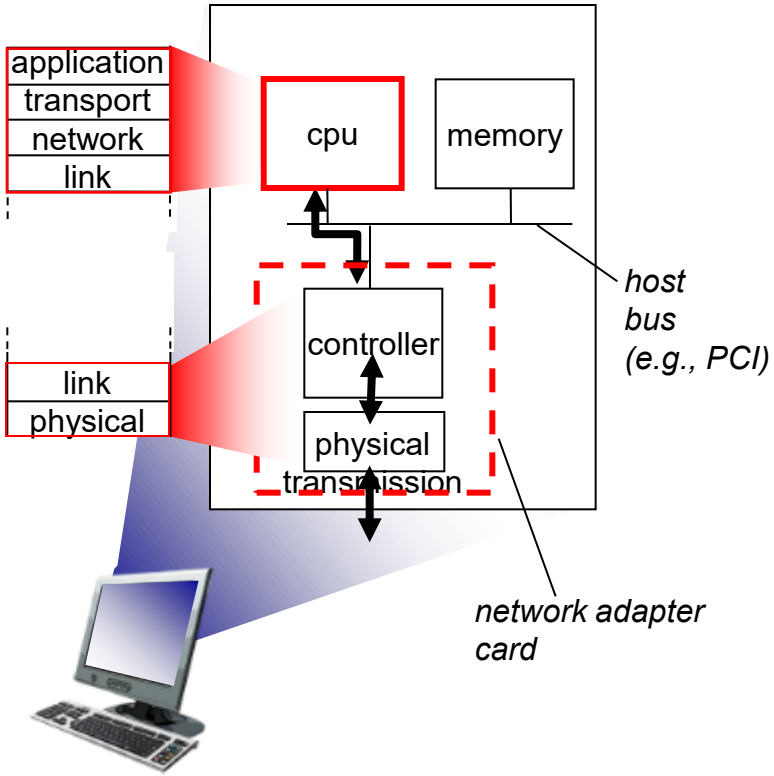
data-link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link

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Where is the link layer implemented?

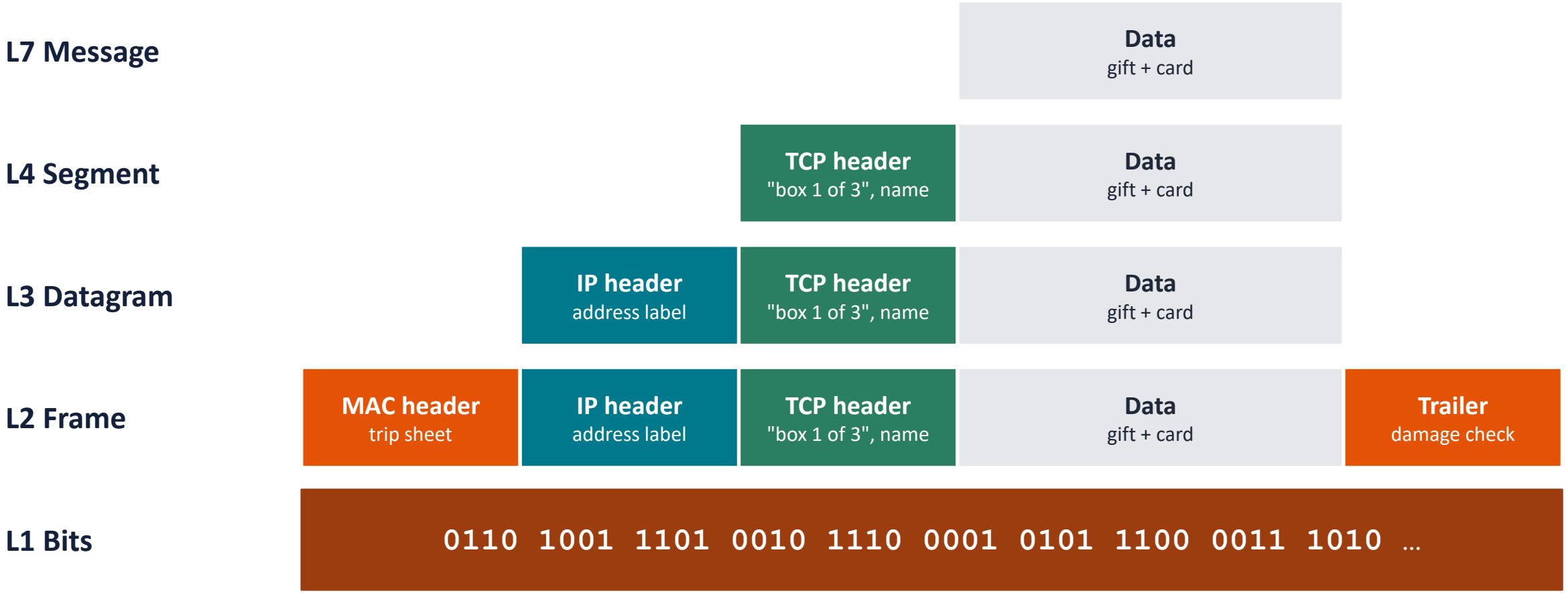
- in each and every host
- link layer implemented in "adaptor" (aka *network interface card* NIC) or on a chip
  - Ethernet card, 802.11 card; Ethernet chipset
  - implements link, physical layer
- attaches into host's system buses
- combination of hardware, software, firmware





Box inside a box: encapsulation

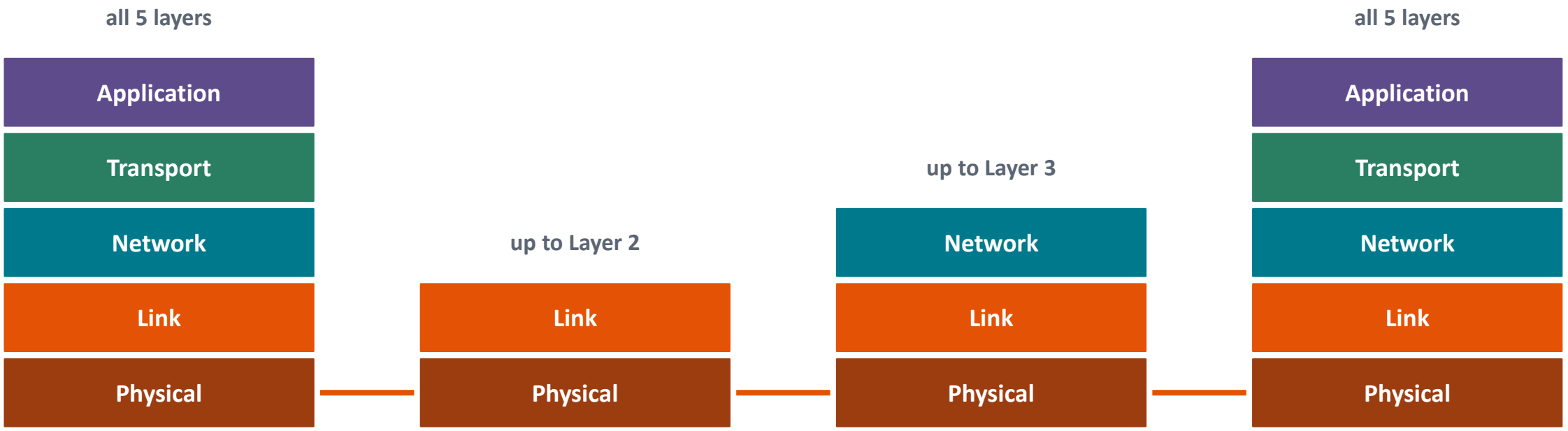
Going down, each layer adds its own label. Going up, each layer removes it.



Layers 5 and 6 are usually handled inside the application, so they are folded into "data" here.

Who opens which box?

Hosts, switches, and routers process different layers



The label stays. The trip sheet changes.

Follow the gift through all 4 truck trips.

Trip	From → To	Address label (IP)	Trip sheet (MAC)
1	Dorm → Bangkok hub	**Dorm → Condo**	**Dorm → Bangkok hub**
2	Bangkok hub → Nakhon Sawan hub	**Dorm → Condo**	**Bangkok hub → Nakhon Sawan hub**
3	Nakhon Sawan hub → Chiang Mai hub	**Dorm → Condo**	**Nakhon Sawan hub → Chiang Mai hub**
4	Chiang Mai hub → Friend's condo	**Dorm → Condo**	**Chiang Mai hub → Friend's condo**



IP = end to end
Source and destination stay the same for the whole journey.



MAC = one hop only
Rewritten at every hub for the next trip.

Where does the analogy break?

Every analogy has limits. Knowing them avoids misconceptions.



Lost without a call

An overloaded router simply drops packets. Nobody phones you to apologize.



Different routes, wrong order

Packets of the same message can take different paths and arrive out of order.



Copied, not carried

Data is not physically moved. Each hop receives signals and re-sends a new copy.



Layers 5–6 are blurry

On the real Internet, applications usually handle these jobs themselves.



Talk with your neighbor for 3 minutes: find one more place where the analogy breaks.

QUICK CHECK · 4 MIN

Match the story to the network

Q1 Splitting the gift into "box 1 of 3" happens at…

- A Layer 3
- B Layer 4
- C Layer 2
- D Layer 7

Q2 The condo front desk that knows each room is…

- A DNS
- B ARP
- C NAT
- D TCP

Q3 Which label is rewritten at every hub?

- A Address label (IP)
- B Trip sheet (MAC)
- C Birthday card (data)
- D None of them

Q4 A switch reads up to which layer?

- A Layer 1
- B Layer 2
- C Layer 3
- D Layer 4

BEFORE THE BREAK

If every router already knows the destination IP address… why do we need MAC addresses at all?



10-minute break

Think about it. We answer it in Block 2.

BLOCK 2

Welcome to the Link Layer



One truck trip at a time

So why do we need MAC addresses?

Three reasons an IP address cannot cover

A fair objection: "Every device on the LAN already has an IP address. Isn't the MAC just redundant?"

An address before any IP

Your laptop joins the Wi-Fi with no IP. The DHCP request must still reach the server, and the reply must return to this one card.

You cannot use an IP address to obtain an IP address.

ARP itself proves it

ARP frames are not IP packets — EtherType 0x0806, not 0x0800. The link also carries 802.1X and IPv6 alongside IPv4.

If delivery needed IP, the protocol that finds IP → MAC could never run.

Nobody has to assign it

A MAC is burned in at the factory, unique worldwide with no coordination. An IP must be handed out by the network owner.

A new card works on any network on Earth before anyone configures anything.

Not a reason: "the NIC filters in hardware". A card could filter on any field at a fixed offset, including an IP. Cheap filtering is a *benefit*, not a justification.

Now the reverse: why not use MAC everywhere and drop IP? Flat addresses cannot be aggregated — a router would need one table row per device on Earth.

MAC is the name an interface has before it has any configuration, and it belongs to the link.
IP is the name the internetwork gives it because of where it sits.

Nodes, links, and frames

Kurose, Chapter 6: the Link Layer and LANs



Node

Any device that runs the link layer: hosts, routers, switches, access points.

Link

A communication channel between two adjacent nodes (orange lines above).

Frame

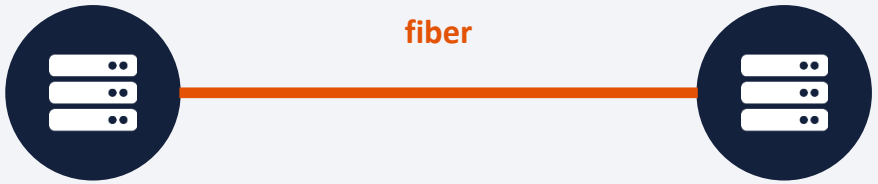
The Layer 2 packet. It carries one datagram across one link.

Like a parcel that goes by motorbike, then truck, then train: each link can use a different Layer 2 protocol.

Two kinds of links

A private road or a shared street?

Point-to-point



One sender and one receiver, like a private road between two hubs.

Examples: a fiber link between two routers, or the cable from your laptop to a switch port.

No sharing problem

Broadcast (shared)



Many nodes share one channel and everyone hears everything, like motorbikes in one busy soi.

Examples: Wi-Fi, cellular, satellite, and old shared-cable Ethernet.

Who talks when? → Class 3

What does the link layer do?

Four services, and where we cover them

Framing

Today + Class 3

Puts the datagram in a frame with a header and trailer.

→ *Attach the trip sheet*

Link access

Class 3

Rules for who may transmit when many nodes share one channel.

→ *Many trucks, one narrow road*

Reliable delivery

Class 3

Some links (like wireless) re-send lost frames right away.

→ *Driver goes back for a dropped box*

Error detection

Class 2 (video)

Receiver checks whether bits were flipped on the way.

→ *Inspect the box at each hub*

A frame up close: Ethernet

Details come in Class 3. For now, notice the shape.



Preamble

Wakes up the receiver and syncs its clock.

Destination first

The NIC can stop reading early if the frame is not for it.

Type

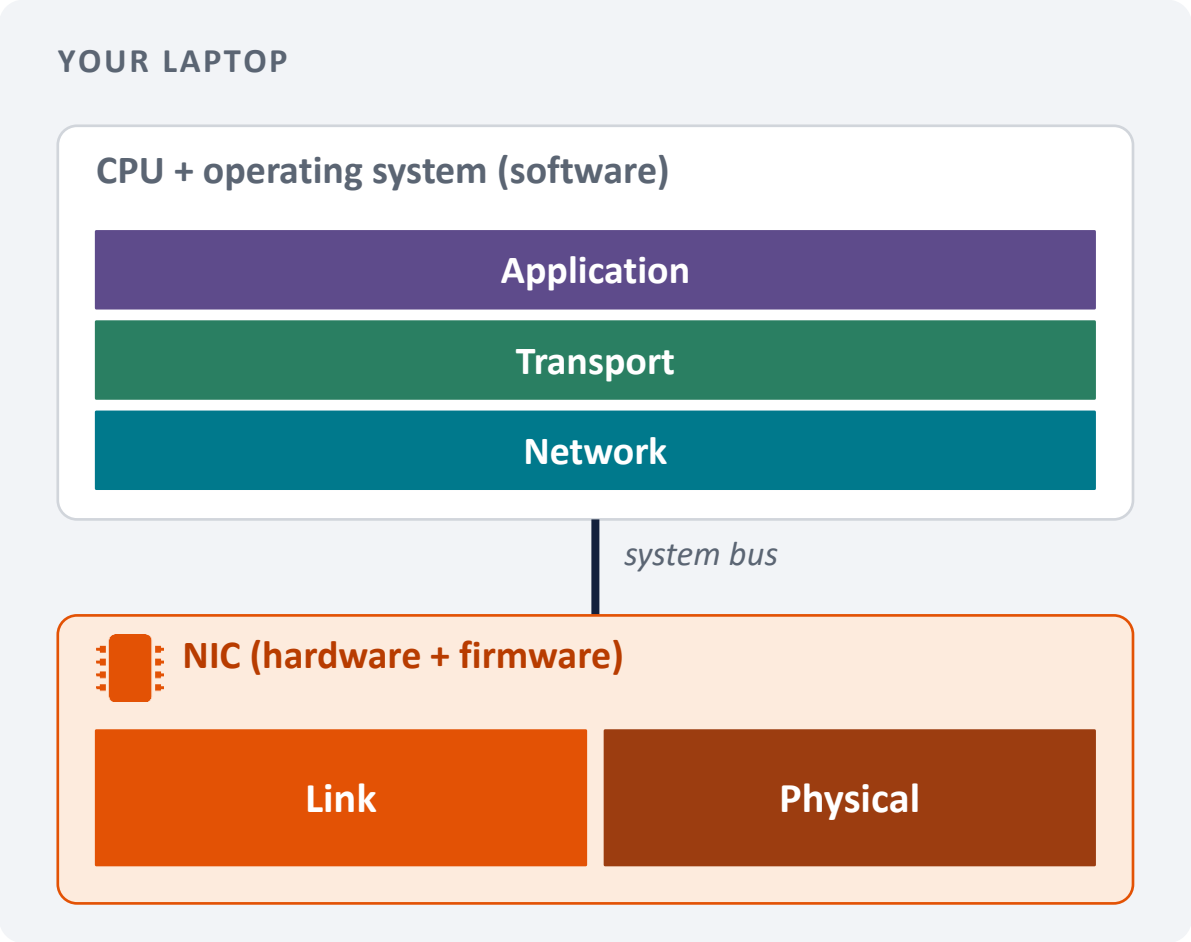
```
0x0800 = IPv4
0x86DD = IPv6
0x0806 = ARP
(Block 3)
```

CRC

The damage check. The sender computes it; the receiver verifies it. Class 2 video.

Where does Layer 2 live?

Mostly in the network interface card (NIC), not in the operating system



The NIC builds frames, checks errors, and handles channel access.

Part of Layer 2 still runs as driver software, for example filling in addresses.

Wi-Fi and Ethernet ports are two separate NICs, each with its own MAC address.

What the two network cards do

Sending side and receiving side of one link

Sending NIC

1. Take a datagram from Layer 3
2. Build the frame: header + error-check bits
3. Follow the access rules of the link
4. Send the bits



Receiving NIC

1. Receive the bits
2. Check for errors
3. Is the destination MAC mine (or broadcast)?
4. Remove the frame; pass the datagram up

Error found, or not my MAC? The frame is dropped silently.

Anatomy of a MAC address

Also called a LAN, physical, or hardware address (example below)

3C:22:FB : **9A:41:07**

First 24 bits: OUI
Identifies the manufacturer (assigned by IEEE)

Last 24 bits
Set by the manufacturer for each device

48 bits

written as 6 bytes in hexadecimal

≈ 281 trillion

possible addresses (2^{48})

FF:FF:FF:FF:FF:FF

broadcast: "to everyone on this LAN"

ACTIVITY · 7 MIN

Find your phone's MAC address



iPhone
Settings → Wi-Fi → tap (i) next to your network → Wi-Fi Address

Android
Settings → Network & internet → your Wi-Fi network → details (menu names vary by brand)

Discuss with your neighbors

1. Compare the first 3 bytes. Same brand, same OUI?
2. Is your address the same on every Wi-Fi network?
3. Why would a phone want to hide its real MAC?



Look closely at the second hex digit
Is it 2, 6, A, or E? Write down your first byte. We explain what it means on the next slide.

Why your phone hides its real MAC

Private (randomized) Wi-Fi addresses



How to spot it

If the second hex digit is 2, 6, A, or E (like 3A:… or 5E:…), the "set locally" bit is on. Software made this address, not the manufacturer.



Why phones do it

Phones announce their MAC while searching for Wi-Fi. A mall or a station could track you. A different random address per network breaks the tracking.



The side effect

Networks that recognize devices by MAC may see your phone as a new device. For network admins, a MAC is no longer a reliable ID.

On by default since iOS 14 and Android 10.

Is this frame for me?

The NIC checks the destination MAC of every frame it hears

DESTINATION MAC	RESULT	TYPE	LIKE A SCHOOL ANNOUNCEMENT
= my own MAC	Accept	Unicast	*"Somchai, please come to the office."*
= FF:FF:FF:FF:FF:FF	Accept	Broadcast	*"Attention, all students."*
= a group I joined (e.g. 01:00:5E:…)	Accept	Multicast	*"Members of the IT club, please…"*
anything else	Drop	—	*Someone else's name: ignore it.*

Two special bits in the first byte

Lowest bit: 0 = individual, 1 = group. Next bit: 0 = set by the manufacturer, 1 = set locally by software.

BLOCK 2 RECAP

Four things to remember

1. A link connects two neighboring nodes, and a frame crosses one link.
2. Layer 2 lives mostly in the network interface card (NIC).
3. A MAC address is a flat 48-bit ID that only matters on one link.
4. IP gets the data to the right network. MAC gets it to the right device on the link.



10-minute break

Next: how does a device find the MAC address it needs?

BLOCK 3

ARP: from IP to MAC



Address Resolution Protocol

The problem: I know the IP. What is the MAC?

Before sending a frame, the NIC must fill in the destination MAC address.

THE FRAME HOST A WANTS TO SEND

Dst MAC	Src MAC	Src IP	Dst IP	Data
? ? ?	3C:22:FB:9A:41:07	192.168.1.10	192.168.1.7	…

Already known

The destination IP (from DNS or the application) and its own addresses.

Missing

The MAC address of the device with IP 192.168.1.7 on this LAN.

ARP answers

"Which MAC belongs to this IP on my subnet?" ARP works only inside one subnet.

Parcel version: the hub knows the next stop is Nakhon Sawan, but which truck at which dock goes there? Shout and ask.

ARP vs DNS: two different translators

Students often mix them up

	DNS	ARP
Question	"Which IP has this name?"	"Which MAC has this IP?"
Example	kmitl.ac.th → its IP address	192.168.1.7 → 58:1C:0A:22:B3:7E
Scope	The whole Internet	One subnet only
Where it runs	Application layer, over UDP	Between L2 and L3, directly in frames
Who answers	DNS servers	The owner of that IP
Cache	Yes	Yes: the ARP table

Think: on a freshly connected laptop, which usually happens first when you open a website?

Inside an ARP message

Carried directly in an Ethernet frame with Type 0x0806, not inside IP

Hardware type 1 = Ethernet		Protocol type 0x0800 = IPv4
HW length 6	**Protocol length** 4	**Operation** 1 = request · 2 = reply
Sender MAC		
Sender IP		
Target MAC all zeros in a request: this is what we want to learn		
Target IP		

Example: request from host A

Operation
`1 (request)`

Sender MAC
`3C:22:FB:9A:41:07`

Sender IP
`192.168.1.10`

Target MAC
`00:00:00:00:00:00`

Target IP
`192.168.1.7`

In the reply, all four addresses are filled in.

ARP in action

Host A (192.168.1.10) looks for 192.168.1.7



Step 1 · ARP query

Broadcast: every host on the LAN receives it

`To: FF:FF:FF:FF:FF:FF`

"Who has 192.168.1.7? Tell 192.168.1.10."



Step 2 · ARP reply

Unicast: only host A receives it

`To: 3C:22:FB:9A:41:07`

"192.168.1.7 is at 58:1C:0A:22:B3:7E."

Step 3 · Host A saves the answer in its ARP table and sends the real frame.

HOST A's ARP TABLE

IP address	MAC address	TTL
192.168.1.1	A4:91:B1:0C:55:20	14 min
192.168.1.7	**58:1C:0A:22:B3:7E**	**20 min**

- Entries expire after a while (TTL), so the table stays up to date.
- Plug-and-play: no administrator needed.
- B also learns A's MAC from the query.

Human ARP

What did we just see?

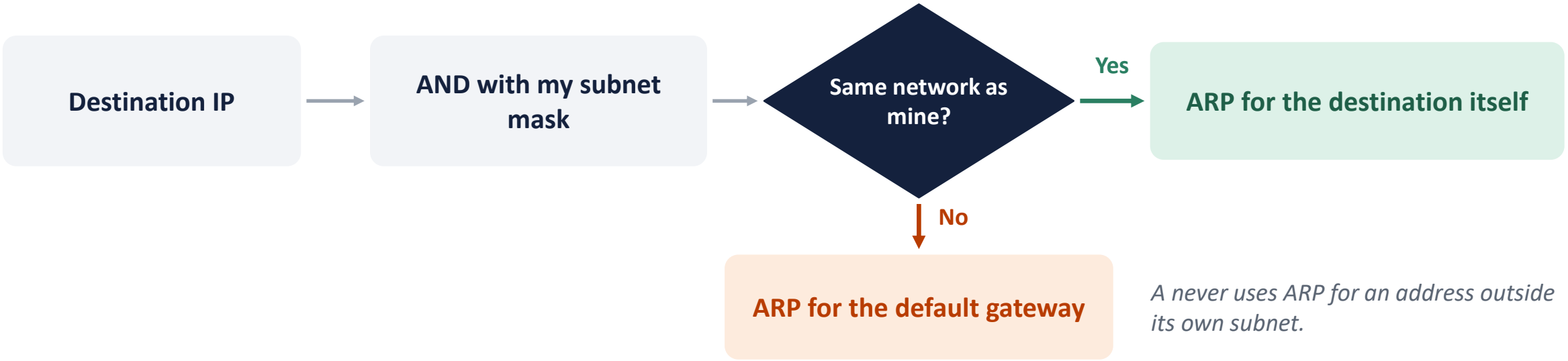
Human ARP debrief

Everyone heard the broadcast	Broadcasts cost every device a little work.
Only the owner answered	Everyone else ignored the question.
No shouting the second time	Caching saves time and traffic.
The owner learned the sender too	The query carried the sender's MAC, so the reply went straight back.
Nobody had 10.0.0.20	It is on another subnet. Ask for the router instead.

Think: what if 1,000 devices shared one LAN? → smaller broadcast domains (VLANs) in Class 3

Same subnet or not? Host A decides

Host A: 192.168.1.10, mask 255.255.255.0, so its network is 192.168.1.0



A never uses ARP for an address outside its own subnet.

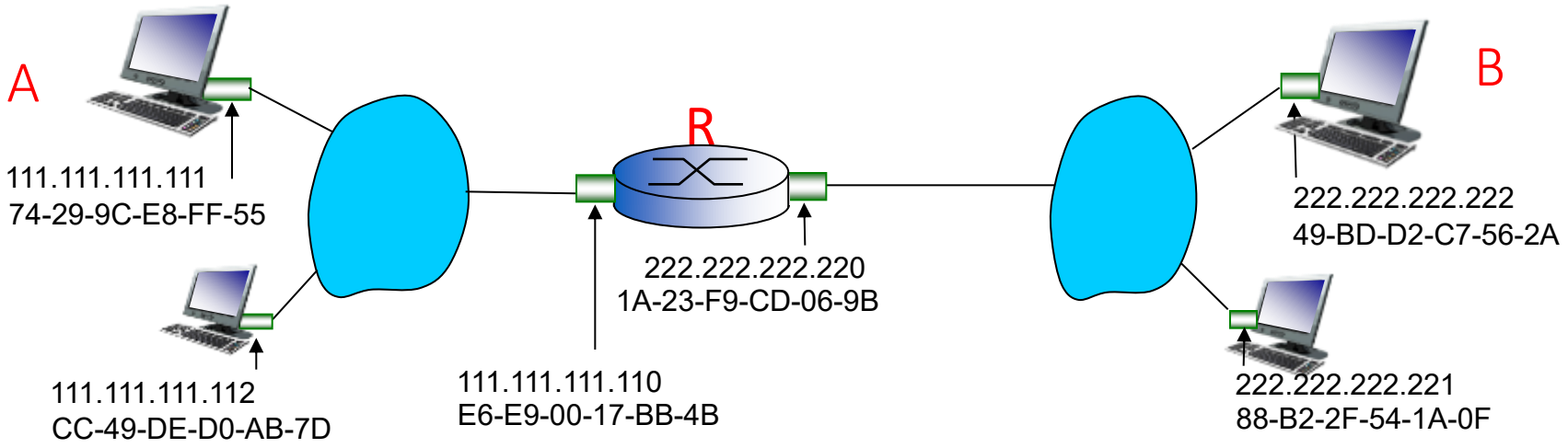
Destination	AND 255.255.255.0	Equals 192.168.1.0?	So A uses ARP for…
`192.168.1.7`	`192.168.1.0`	Yes	`192.168.1.7`
`10.0.0.20`	`10.0.0.0`	No	`192.168.1.1 (gateway)`

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Addressing: routing to another LAN

walkthrough: send datagram from A to B via R

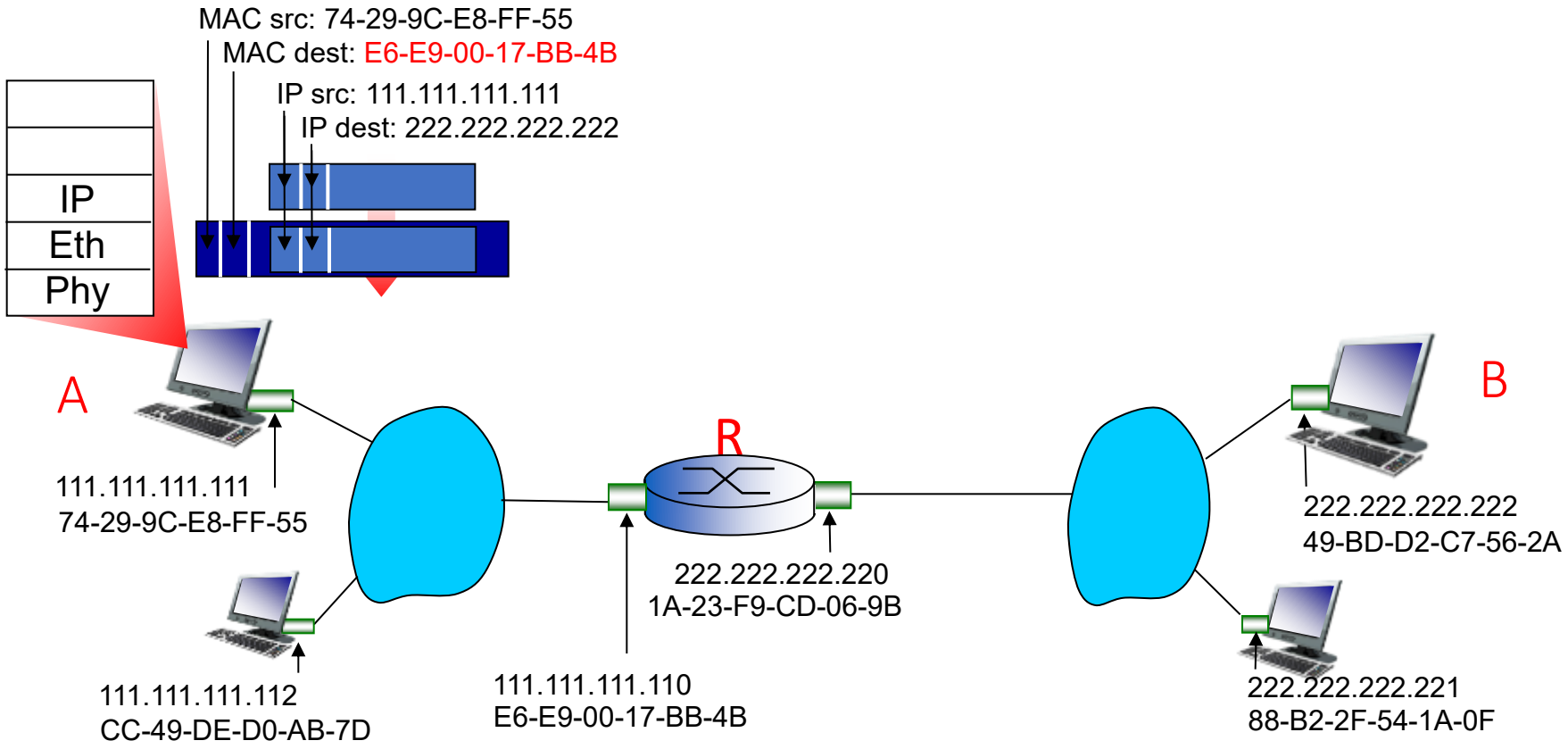
- focus on addressing – at IP (datagram) and MAC layer (frame)
- assume A knows B's IP address
- assume A knows IP address of first hop router, R (how?)
- assume A knows R's MAC address (how?)



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Addressing: routing to another LAN

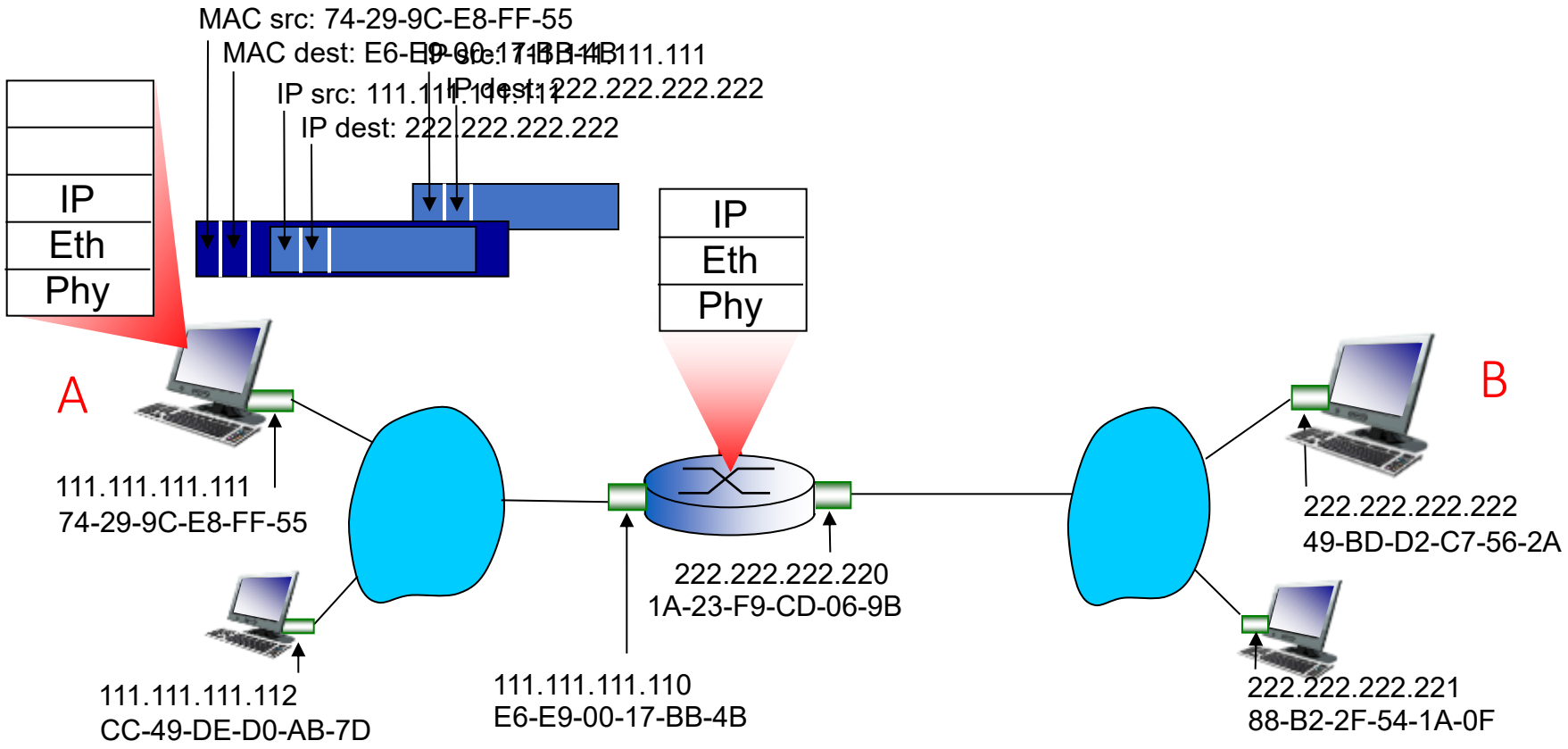
- A creates IP datagram with IP source A, destination B
- A creates link-layer frame with R's MAC address as destination address, frame contains A-to-B IP datagram



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Addressing: routing to another LAN

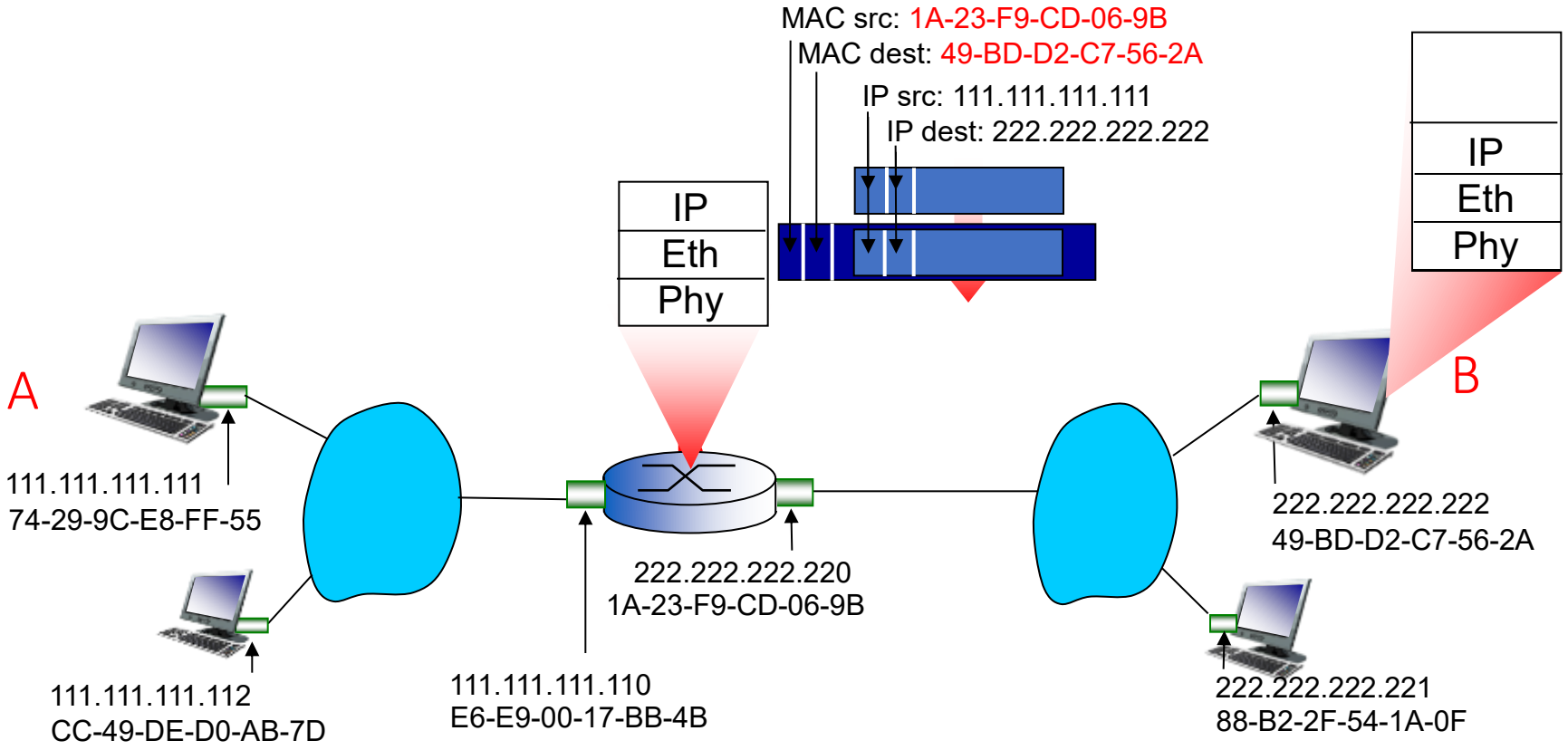
- frame sent from A to R
- frame received at R, datagram removed, passed up to IP



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Addressing: routing to another LAN

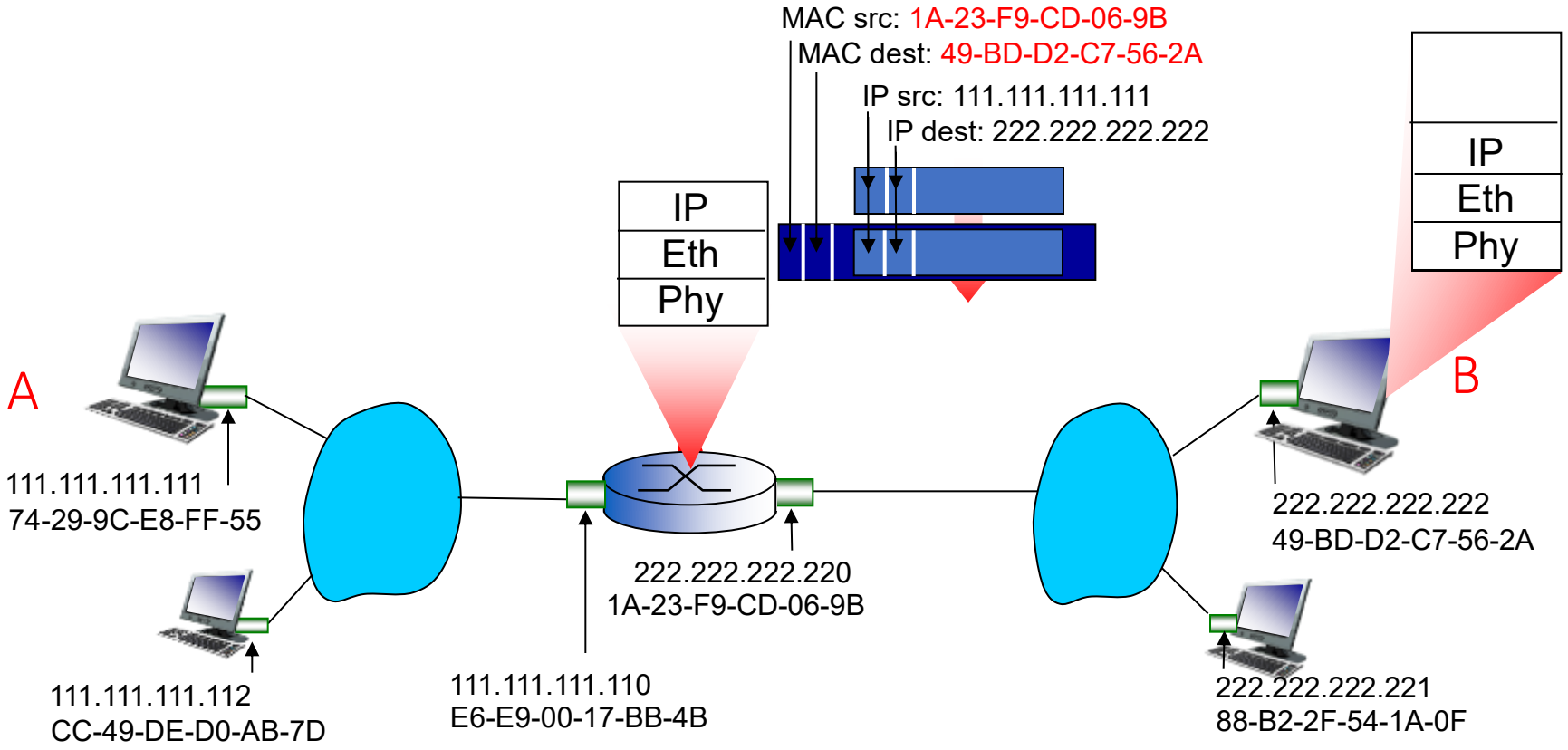
- R forwards datagram with IP source A, destination B
- R creates link-layer frame with B's MAC address as destination address, frame contains A-to-B IP datagram



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Addressing: routing to another LAN

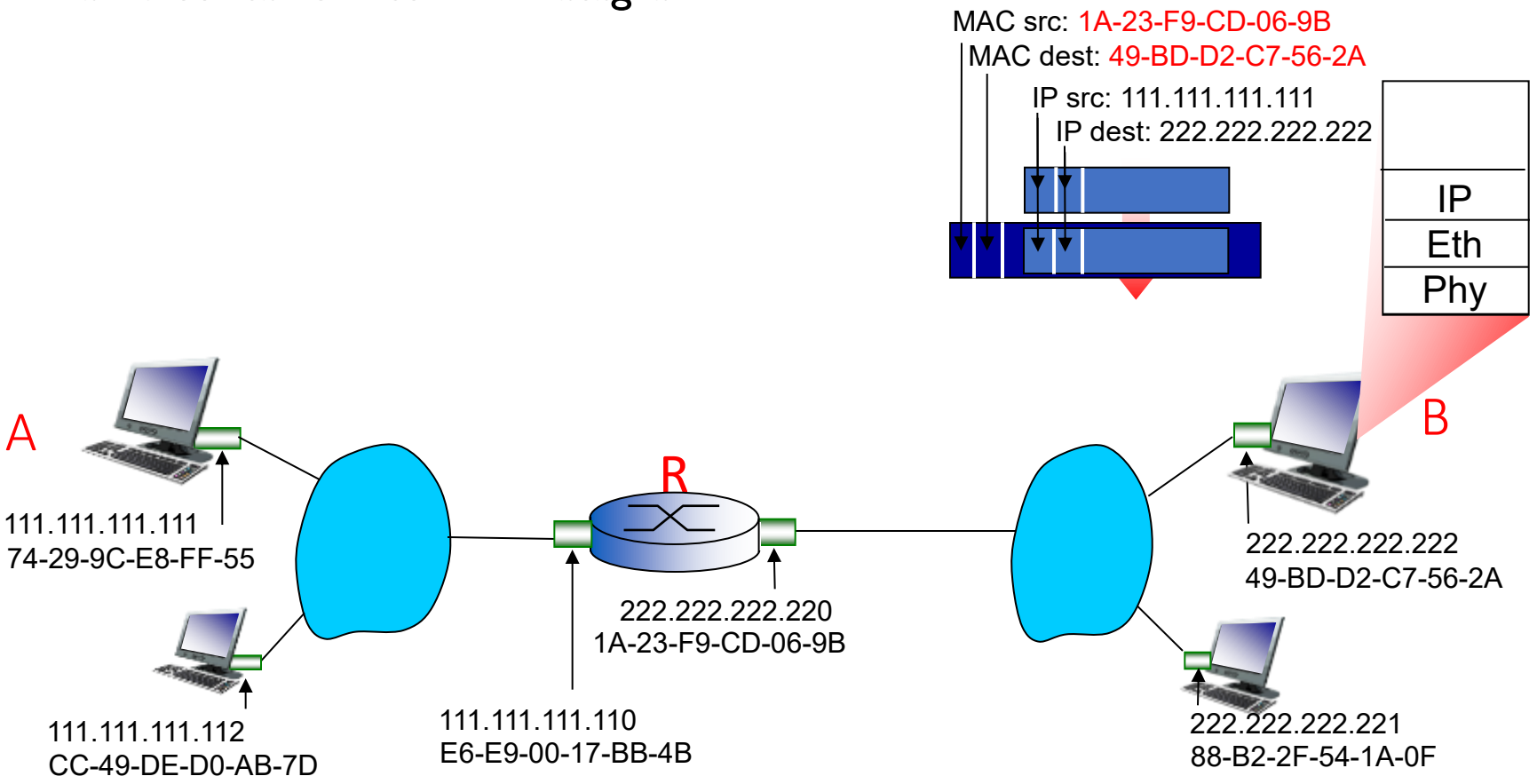
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Addressing: routing to another LAN

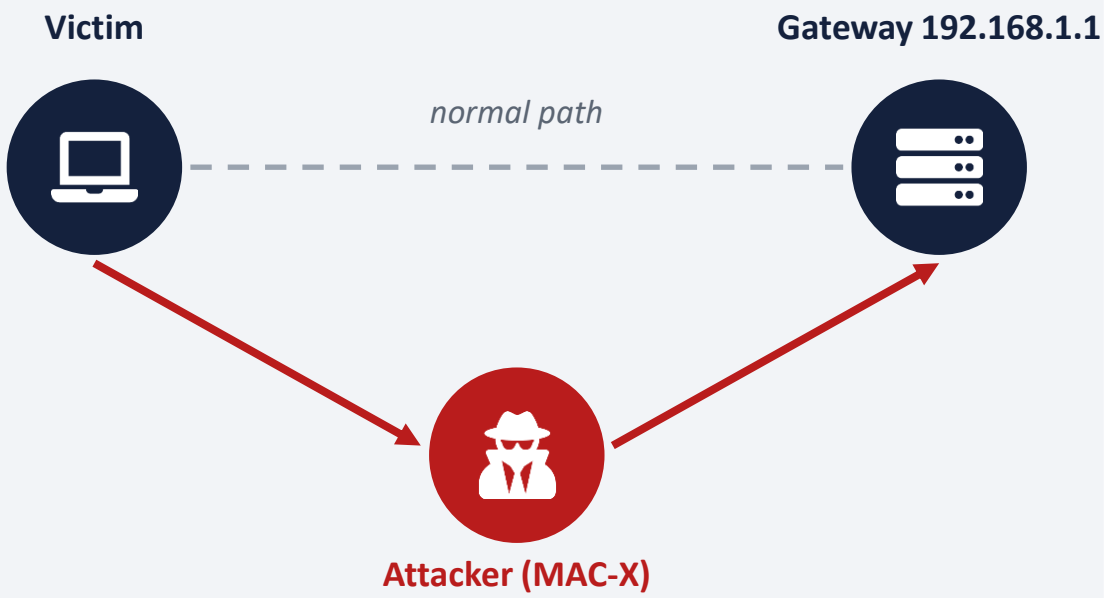
- R forwards datagram with IP source A, destination B
- R creates link-layer frame with B's MAC address as destination address, frame contains A-to-B IP datagram



* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

ARP has no lock: ARP spoofing

Why the protocol we just learned is a security concern



Fake reply to the victim: "192.168.1.1 is at MAC-X"

Why it works

ARP trusts any reply, even one nobody asked for. There is no authentication.

Defenses

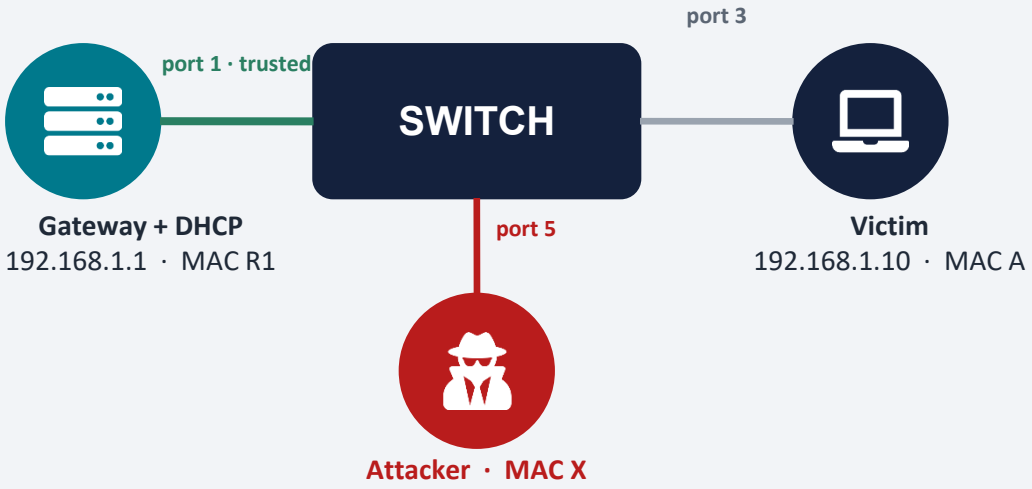
- Switches check ARP against a trusted list (Dynamic ARP Inspection + DHCP snooping)
- Static ARP entries for critical devices
- Encryption (TLS, VPN) so intercepted data is unreadable

Trying this on any network without permission, including the campus network, is illegal.

Stopping ARP spoofing: the switch keeps a register

DHCP snooping writes down who got which address. Dynamic ARP Inspection checks every ARP against it.

STEP 1 · THE SWITCH WATCHES DHCP



STEP 2 · IT BUILDS A BINDING TABLE

IP address	MAC	Port
192.168.1.1	R1	1 (trusted)
192.168.1.10	A	3
192.168.1.66	X	5

The switch saw the DHCP server hand out every one of these, so this table is the truth.

STEP 3 · EVERY ARP MESSAGE IS CHECKED AGAINST THE TABLE

Real host on port 3

`"I am 192.168.1.10, my MAC is A"` → Match → forward it

Table says 192.168.1.10 = A on port 3.

Attacker on port 5

`"I am 192.168.1.1, my MAC is X"` → Mismatch → drop, log, disable the port

Table says 192.168.1.1 = R1 on port 1.

Like the condo front desk from earlier: it keeps a register of who lives in which room, so it can catch anyone claiming to be room 101.

A day in the life of a web request

Back to the warm-up question, now with every layer in place

1



Join Wi-Fi

DHCP gives an IP. With no IP yet, the laptop broadcasts using its MAC.

L7 · DHCP

2



Find router

ARP gets the gateway's MAC address.

L2/L3 · today

3



Find server

DNS turns kmitl.ac.th into an IP address.

L7 · DNS

4



Connect

TCP three-way handshake with the web server.

L4 · TCP

5



Secure

TLS agrees on encryption keys.

L6 · TLS

6



Get page

HTTP request and response.

L7 · HTTP

Under every step: each packet is framed on every link (Layer 2) and sent as signals (Layer 1).

Based on Kurose, "A Day in the Life of a Web Page Request" (end of Chapter 6)

Today's five takeaways

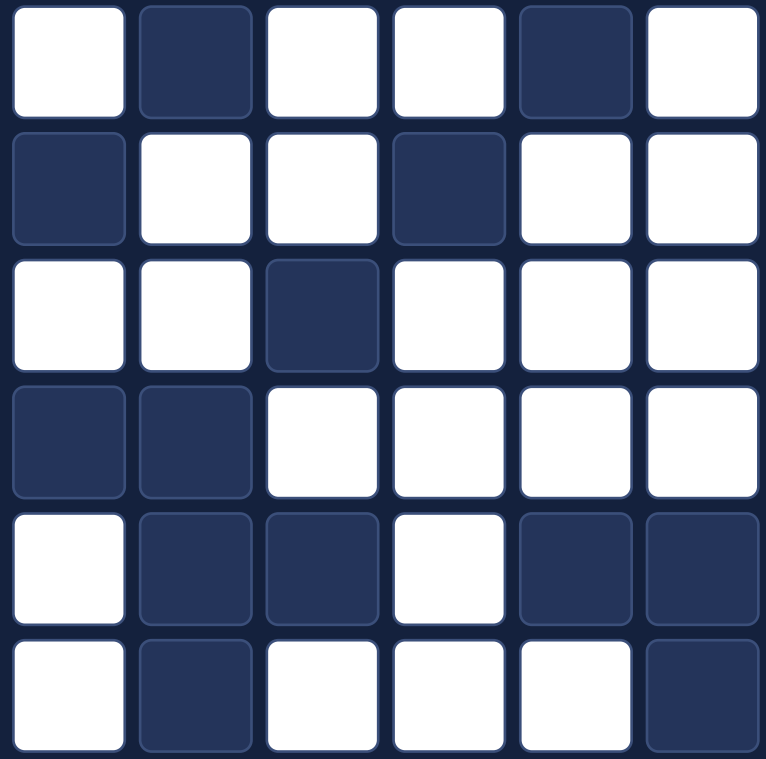
If you remember only one slide, remember this one

1	**Encapsulation**	Each layer adds its own header on the way down and removes it on the way up.
2	**Who reads what**	Hosts process all layers, routers up to Layer 3, switches up to Layer 2.
3	**IP stays, MAC changes**	IP is end to end. MAC covers one hop only.
4	**MAC address**	A flat 48-bit ID. FF:FF:FF:FF:FF:FF means everyone on the LAN.
5	**ARP**	Maps IP to MAC inside a subnet. For other subnets, use ARP for the gateway.

NEXT: CLASS 2 · RECORDED VIDEO

How does the receiver know a bit was flipped?

Error detection: parity, checksum, and CRC. Watch the video and finish the worksheet before our next live class.



Someone flipped one card. Can you find it? The trick is in the video.